# Naive bayes

$$P(A \text{ and } B) = P(A) \times P(B/A)$$

$$P(A \text{ and } B) = P(B \text{ and } A)$$

$$P(A) \times P(B/A) = P(B) \times P(A/B)$$

# $$P(B/A) = \frac{P(B) \times P(A/B)}{P(A)}$$

$$\underbrace{X_1 \quad X_2 \quad X_3}_{A} \quad \underbrace{Y}_{B} \begin{cases} \text{yes} \\ \text{No} \end{cases}$$

$$P\left(Y/_{X_1 X_2 X_3}\right) = \frac{P(Y) \times P(X_1/_Y)\, P\left(\frac{X_2}{Y}\right) P(X_3/_1)}{P(X_1)\, P(X_2)\, P(X_3)}$$

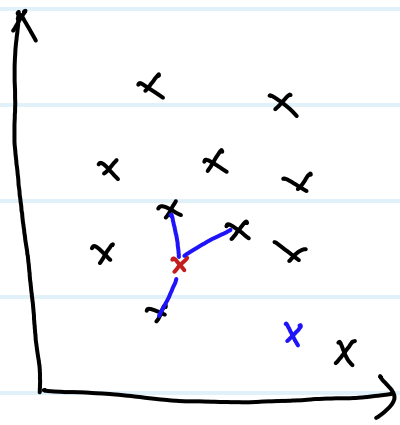
$$P\left(\frac{\text{yes}}{X_1 X_2 X_3}\right) = \frac{P(\text{yes}) \times P\left(\frac{X_1}{\text{yes}}\right) P\left(\frac{X_2}{\text{yes}}\right) P\left(\frac{X_3}{\text{yes}}\right)}{\cancel{P(X_1)\, P(X_2)\, P(X_3)}}$$

$$P\left(\frac{No}{x_1 x_2 x_3}\right) = \frac{P(No) \times P\left(\frac{x_1}{No}\right) P\left(\frac{x_2}{No}\right) P\left(\frac{x_3}{No}\right)}{\cancel{P(x_1)\ P(x_2)\ P(x_3)}}$$

# KNN

$K \approx [3, 5, 7, 9, 11]$

$K = 3$



Distance
(i) euclidean dist.
(ii) manhattan dist.

(i) euclidean $\rightarrow$ $D = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2}$

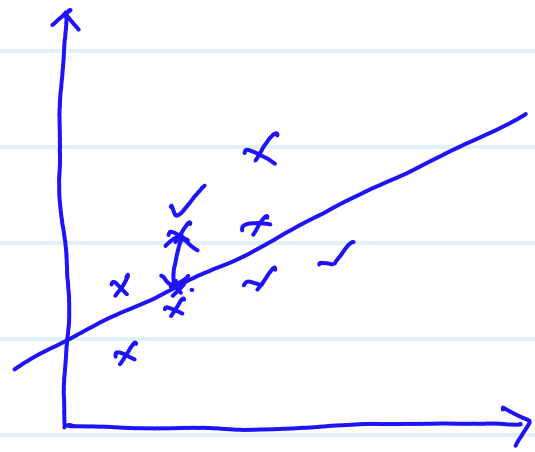
| X | Y | | Dist 6 |
|---|---|---|---|
| 1 | 2 | $\rightarrow$ | 3 |
| 2 | 4 | $\rightarrow$ | 4 |
| 3 | 7 ✓ | $\rightarrow$ | 2 |
| 4 | 1 ✓ | $\rightarrow$ | 1 |
| 5 | 3 ✓ | $\rightarrow$ | 1.5 |
| 6 | ? 3.66 | | |

$$\frac{7+1+3}{3} = \frac{11}{3}$$

② manhatten $= |X_2 - X_1| + |Y_2 - Y_1|$

# Linear Regression

① Linearity
② homoscedacity
③ multicolinearity
④ Normality



$$Y = mx + c$$

or

$$Y = \theta_0 + \theta_1 X_1$$

$$Y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3$$

$\theta_0, \theta_1$ new $x_1$ ✓ $= \hat{Y}$

$$\text{Loss} = (Y - \hat{Y})^2$$

$$\text{cost function} = \frac{1}{2n} \sum_{i=1}^{n} (Y_i - \hat{Y})^2$$

$$= \frac{1}{2n} \sum_{i=1}^{n} \left[ Y_i - (\theta_0 + \theta X_1) \right]^2$$

# Repeat conversion theorem

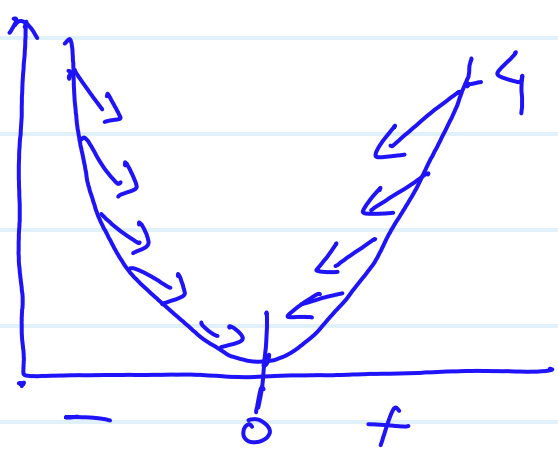
$\theta_0$ and $\theta_1$

$$\theta_{0_{new}} = \theta_{0_{old}} - \alpha \frac{\partial L}{\partial \theta_{0_{old}}}$$

$$\theta_{1_{new}} = \theta_{1_{old}} - \alpha \frac{\partial L}{\partial \theta_{1_{old}}}$$

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^{n} \left[ y_i - (\theta_0 + \theta_1 x_1) \right]^2$$

Gradient decent




$-4 - (-2)$

$-2$

$4 - (+2)$

$2$

$$L_1 = \text{cost funct} + \lambda \sum |\theta_1|$$

$$L_2 = \text{cost funct} + \lambda \sum (\theta_1)^2$$

$$EN = \text{cost func} + \lambda \sum |\theta_1| + \lambda \sum (\theta_1)^2$$

# VIF $\Rightarrow \dfrac{1}{1+R^2}$

VIF > 5

# Evaluation metrics

① $\underline{m}SE = \dfrac{\sum (Y-\hat{Y})^2}{n}$

② $MAE = \dfrac{\sum |Y-\hat{Y}|}{n}$

③ $RMSE = \sqrt{MSE}$

④ $\underline{R^2}score = 1 - \dfrac{SS_R}{SS_T} \quad \dfrac{(RSS)}{(TSS)}$

$$= 1 - \frac{\sum (Y-\hat{Y})^2}{\sum (Y-\bar{Y})^2}$$

⑤ Adj $R^2$score $= \dfrac{1-(1-R^2)(N-1)}{N-P-1}$

# Logistic Regression

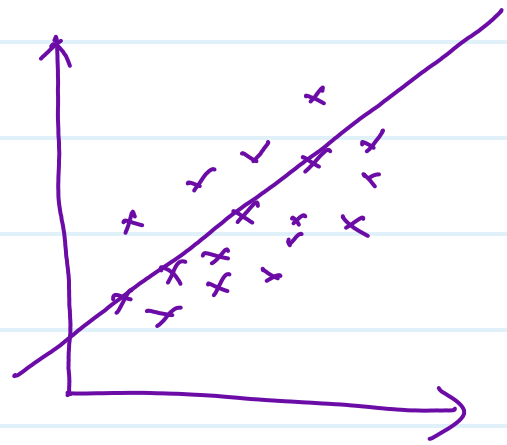
$$Y = mx + c$$

$$Y = \theta_0 + \theta_1 x_1$$

$$Y = Z$$

$$Z = \theta_0 + \theta_1 x_1$$

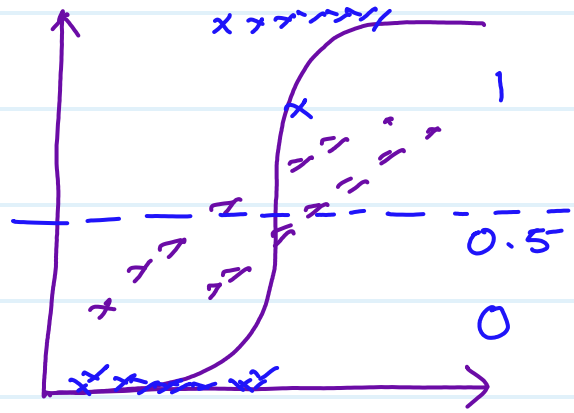
continous. (



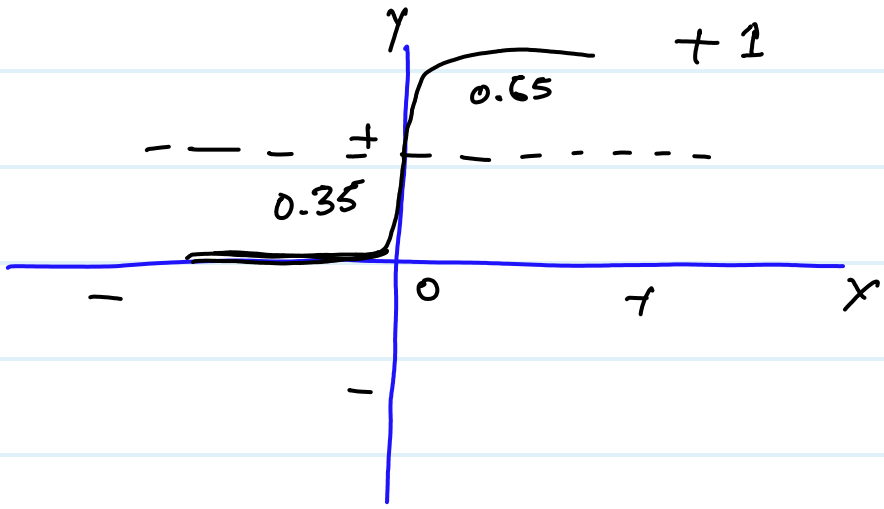
✓ Sigmoid Activation func

$$\sigma = \frac{1}{1 + e^{-Z}}$$

$$= \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1)}}$$




<u>Probability</u> (0 to 1)



linear cost func. (non-convex)
it creates local minima

log loss cost func (convex)

$$= -y\log(y) - (1-y)\log(1-y)$$

$$\Rightarrow \underbrace{-y\log(y)} - \underbrace{(1-y)\log(1-y)}$$

# Performance Metrics –

| | | Act T | Act F |
|---|---|---|---|
| Pred | T | TP | FP |
| | F | FN | TN |

① Confusion metrics

② Accuracy $= \dfrac{TP + TN}{TP + FP + TN + FN}$

③ Precision $= \dfrac{TP}{TP + FP}$

④ Recall $= \dfrac{TP}{TP + FN}$

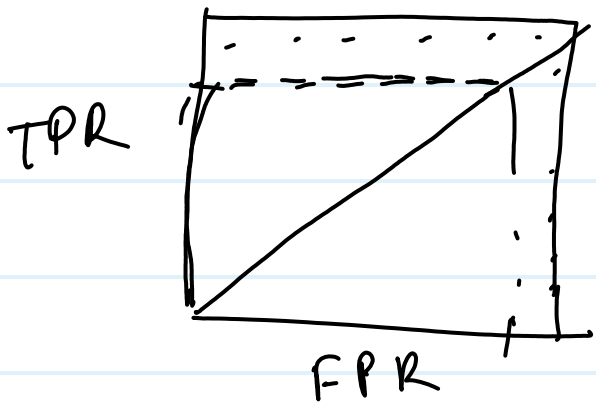
⑤ f1 score $= \dfrac{2 \times (P \times R)}{(P + R)}$

⑥ ROC & AUC

$\Rightarrow$

$$TPR = \frac{TP}{(TP + FN)}$$

$$FPR = \frac{FP}{(FP + TN)}$$




# Decision Tree

① ID3 → entropy, information gain

② CART → Gini impurity

entropy $= -P(Y)\log P(Y) - P(N)\log P(N)$

Infor. gain $= \underbrace{H(S)}_{} - \underbrace{\sum \frac{|S_v|}{S} H(S_v)}_{}$

$=$

entropy $= 0$ to $1$

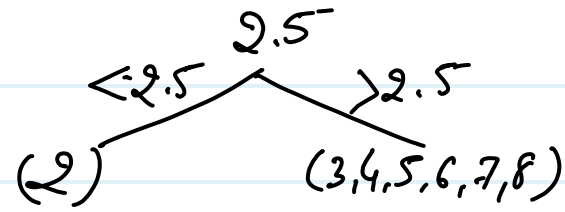
Gini $= 0$ to $0.5$

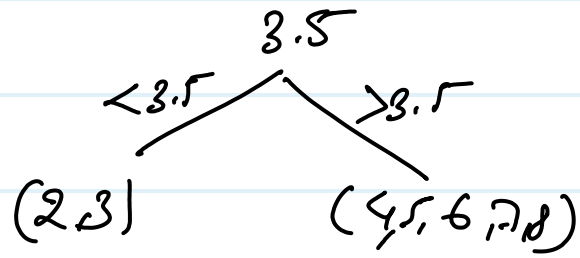
$$\text{Gini} = 1 - P(Ye)^2 - P(No)^2$$

$$=$$

| $X_1$ | $X_2$ | $X_3$ | $X_4$ |
|---|---|---|---|

| $X_1$ | | |
|---|---|---|
| 2 | | Y |
| | 2.5 | |
| 3 | | Y |
| | 3.5 | |
| 4 | | Y |
| | 4.5 | |
| 5 | | N |
| | 5.5 | |
| 6 | | Y |
| | 6.5 | |
| 7 | | N |
| | 7.5 | |
| 8 | | N |

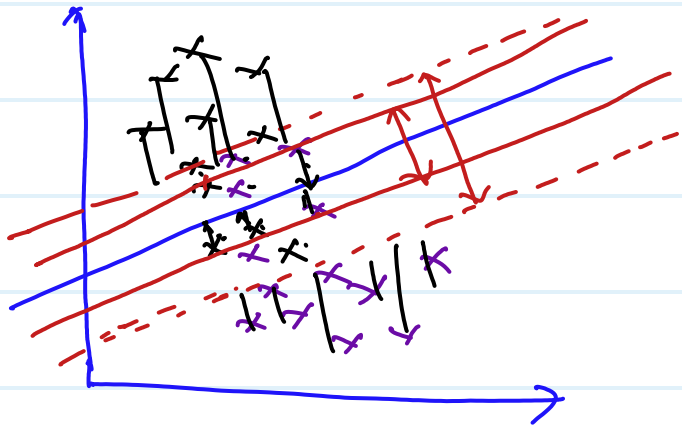



**Parameter** – **Pruning**
- Post pruning
- Pre pruning

I) CCP
II) max_sample_split
III) max_depth
IV) min_sample_leaf
V)

# SVM



$$y = mx + c$$

$$= m_1 x_1 + m_2 x_2 + c$$

$$= w_1 x_1 + w_2 x_2 + b \qquad \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

Transpose $\Rightarrow w^T x + b = \underline{\underline{\text{value}}}$

$$\frac{w^T x + b}{\|w\|} = \checkmark$$

$$w^T x + b > 1 \qquad +1$$

$$w^T x + b < 1 \qquad -1$$

$$a\tilde{x} + b\tilde{y} + c =$$

$$by = -ax - c$$

$$y = \underbrace{-\frac{a}{b}}_{w} x - \underbrace{\frac{c}{b}}_{b}$$

$$w^T x_1 + \cancel{b} = 1$$

$$- \; w^T x_2 + \cancel{b} = \mp 1$$

$$\frac{w^T(x_1 - x_2)}{\|w\|} = \frac{2}{\|w\|}$$

$$= \frac{\|w\|}{2} + \boxed{C_1 \sum \xi_i} \quad \underline{\text{hinge loss}}$$

$$= \frac{\|w\|}{2} + \boxed{C_1 \sum |\xi_1|}$$

<u>Kernel</u>

1. Linear
2. polynomial
3. Radial
4. sigmoid